

DIGITAL IMAGE PROCESSING LABORATORY

Semester	B.E. Semester VII
Subject	Digital Image Processing
Subject Professor In-charge	Prof. Rajashree Soman
Laboratory	Lo5

Student Name	Amey Pawar	Division	B
Roll Number	23108B0057	Batch	2
Grade and Subject Teacher's Signature			

Experiment Number	6
Experiment Title	Image Segmentation – Edge Detection and Thresholding
Aim	To implement and compare classical edge-detection operators (Roberts, Prewitt, Sobel, Laplacian) and thresholding methods (global fixed-value, Otsu, adaptive) for segmenting the iris/pupil region in grayscale near-infrared eye-crop images from the driver-drowsiness dataset, and to quantitatively evaluate object/background separation, boundary quality and noise so as to identify the best-suited edge operator and thresholding method for this dataset. Note on input substitution: the experiment brief specifies using "the best output obtained from Task 5" as input here. Task 5 was not carried out as a separate exercise in this project, so its role is substituted with the best-performing enhancement output from Task 3 (Image Enhancement) instead – specifically histogram equalization (<code>cv2.equalizeHist</code>), which was measured in Task 3 to win the contrast-gain-per- noise-gain comparison on 2 of the 3 test images (see <code>results/task3_metrics.csv</code> and <code>docs/task3_report.pdf</code>). The same 3 dataset images used in Task 3 are reused here so the enhancement-to-segmentation pipeline is directly traceable end to end.
Resources / Apparatus Required	Software: Python 3.11 • OpenCV • NumPy • Matplotlib • scikit-image • Pillow • Jupyter Notebook / VS Code
Algorithm	A. Edge Detection 1. Start the program. 2. Read the enhanced (histogram-equalized) grayscale eye-crop image as a NumPy <code>uint8</code> array. 3. For Roberts, convolve the image with the 2x2 kernels <code>Gx=[[1,0],[0,-1]]</code>, <code>Gy=[[0,1],[-1,0]]</code> via <code>cv2.filter2D</code>. 4. For Prewitt, convolve with the 3x3 kernels <code>Gx=[[-1,0,1],[-1,0,1],[-1,0,1]]</code>, <code>Gy=[[-1,-1,-1],[0,0,0],[1,1,1]]</code> via <code>cv2.filter2D</code>. 5. For Sobel, compute the center-weighted 3x3 gradients with the OpenCV builtin <code>cv2.Sobel(img, CV_64F, 1, 0) / cv2.Sobel(img, CV_64F, 0, 1)</code>.

6. For **Laplacian**, compute the isotropic second derivative with the OpenCV builtin `cv2.Laplacian(img, CV_64F)` and take its absolute value.
7. For every first-derivative operator, compute gradient magnitude `sqrt(Gx^2 + Gy^2)`; normalize every magnitude map to its own peak and cast to `uint8` [0,255] so operators are visually and numerically comparable.
8. Display the original, enhanced and all 4 edge maps side by side.

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9. Stop the program.

B. Thresholding

1. Start the program.
2. Read the enhanced (histogram-equalized) grayscale eye-crop image.
3. For **global thresholding**, apply one fixed cutoff (`thresh=127`) via `cv2.threshold(img, thresh, 255, THRESH_BINARY_INV)` – dark-foreground convention, since the pupil/iris is the darkest structure in these IR crops.
4. For **Otsu thresholding**, reuse `src.segment.otsu_threshold(THRESH_BINARY_INV + THRESH_OTSU)`: the split value is derived automatically from the image's own histogram by maximizing between-class variance, not hand-picked.
5. For **adaptive thresholding**, reuse `src.segment.adaptive_threshold(Gaussian-weighted local mean, 15x15 block, C=3)`: the threshold varies per pixel neighbourhood rather than globally.
6. For every resulting binary mask, run 8-connectivity connected-component labelling (`cv2.connectedComponentsWithStats`) to count components and isolated single-pixel components (noise proxy).
7. Compute Intersection-over-Union of the global and adaptive masks against the Otsu mask (reference) on the same image.
8. Display the enhanced image and all 3 binary masks side by side.
9. Stop the program.

Program code

Trimmed to the essential logic. Full module: `src/edge_segmentation.py`. Every function takes/returns a single-channel `uint8` NumPy array; thresholding wrappers reuse `src/segment.py` (Task 3/Week-3 module) rather than duplicating its Otsu/adaptive logic.

A. Edge Detection

```
ROBERTS_GX = np.array([[1, 0], [0, -1]], dtype=np.float64)
ROBERTS_GY = np.array([[0, 1], [-1, 0]], dtype=np.float64)
PREWITT_GX = np.array([[-1,0,1], [-1,0,1], [-1,0,1]],
dtype=np.float64)
PREWITT_GY = np.array([[-1,-1,-1], [0,0,0], [1,1,1]],
dtype=np.float64)

def _normalize_mag(mag):
    # scale to uint8
    [0,255]
    peak = mag.max()
    if peak <= 0: return np.zeros_like(mag, dtype=np.uint8)
    return np.clip(mag / peak * 255.0, 0, 255).astype(np.uint8)

def roberts_edge(img):
    img_f = img.astype(np.float64)
    gx = cv2.filter2D(img_f, cv2.CV_64F, ROBERTS_GX)
```

```
gy = cv2.filter2D(img_f, cv2.CV_64F, ROBERTS_GY)
return _normalize_mag(np.sqrt(gx**2 + gy**2))
```

```
def prewitt_edge(img):
    img_f = img.astype(np.float64)
    gx = cv2.filter2D(img_f, cv2.CV_64F, PREWITT_GX)
    gy = cv2.filter2D(img_f, cv2.CV_64F, PREWITT_GY)
    return _normalize_mag(np.sqrt(gx**2 + gy**2))
```

```
def sobel_edge(img, ksize=3):
    gx = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=ksize)
    gy = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=ksize)
    return _normalize_mag(np.sqrt(gx**2 + gy**2))
```

```
def laplacian_edge(img, ksize=3):
    lap = cv2.Laplacian(img, cv2.CV_64F, ksize=ksize)
    return _normalize_mag(np.abs(lap))
```

B. Thresholding

```
def global_threshold(img, thresh=127):          # dark-foreground
    convention
    _, mask = cv2.threshold(img, thresh, 255,
cv2.THRESH_BINARY_INV)
    return mask
```

```
def otsu_wrapper(img):                          # reuses
src.segment (Task 3)
    _, mask = otsu_threshold(img)
    return mask
```

```
def adaptive_wrapper(img, block_size=15, C=3):  # reuses
src.segment (Task 3)
    return adaptive_threshold(img, method="gaussian",
block_size=block_size, C=C)
```

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Edge-map binarization for metrics only (figures show the raw magnitude maps): Otsu thresholding was tried first to binarize the gradient-magnitude maps for the component/noise counts, and rejected – measured directly on `good_lighting`, Otsu picked value 13 for Roberts but 91 for Sobel/Prewitt (91 sits past that map's own 99th percentile, 42), collapsing those masks to 1-2 stray pixels. Gradient-magnitude histograms are strongly unimodal (a huge near-zero bulk plus a thin edge tail), which is exactly the shape Otsu is not designed for. A single fixed cutoff (≥ 25 , on each operator's own peak-normalized `[0,255]` map) was used instead for every operator/image, keeping the ratio comparable across operators; see `scripts/make_task6_figures.py`.

Output

Same 3 test images as Task 3 (see `data/samples/manifest.csv` fields `glasses/lighting/reflections`), each loaded, upscaled with `cv2.INTER_NEAREST` (minimum 300px side), and histogram-equalized before edge detection / thresholding.

Label	File	Condition	Native size
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good_lighting	alert/s0031_00366_1_0_1_0_1_02.png	open eye, no glasses, good lighting, no reflection	128x128
poor_lighting	alert/s0014_08297_0_0_1_1_0_02.png	open eye, no glasses, poor/uneven lighting	130x130
glasses_reflection	alert/s0035_00133_0_1_1_2_1_02.png	open eye, eyeglasses + strong IR reflection (glare hotspot)	144x144

Edge detection, per image

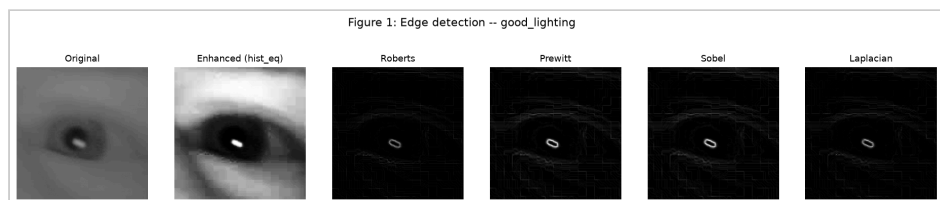


Figure 1: Edge detection (Roberts / Prewitt / Sobel / Laplacian) – good_lighting.

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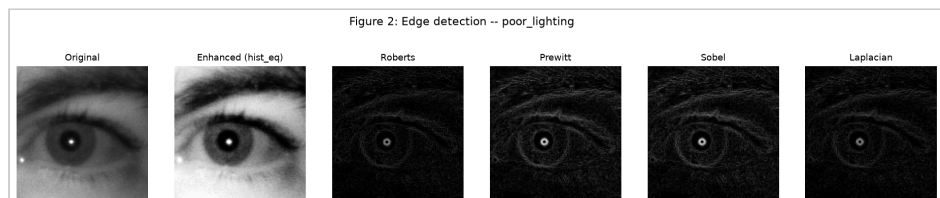


Figure 2: Edge detection (Roberts / Prewitt / Sobel / Laplacian) – poor_lighting.

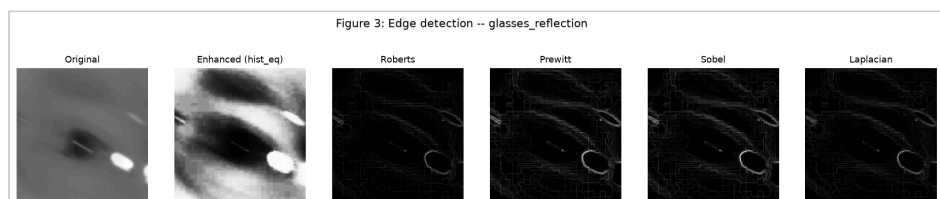


Figure 3: Edge detection (Roberts / Prewitt / Sobel / Laplacian) – glasses_reflection.

Thresholding, per image

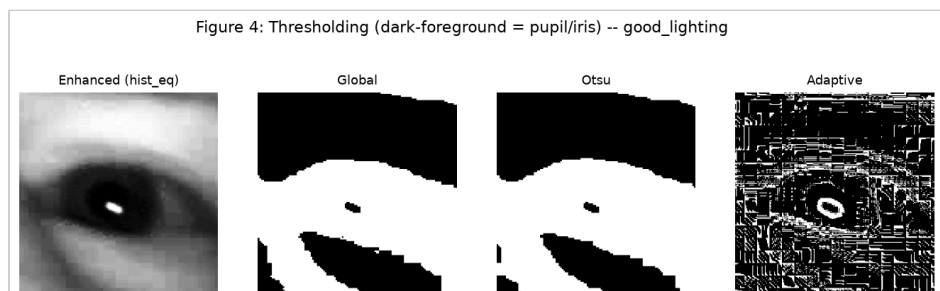


Figure 4: Global / Otsu / adaptive thresholding – good_lighting.

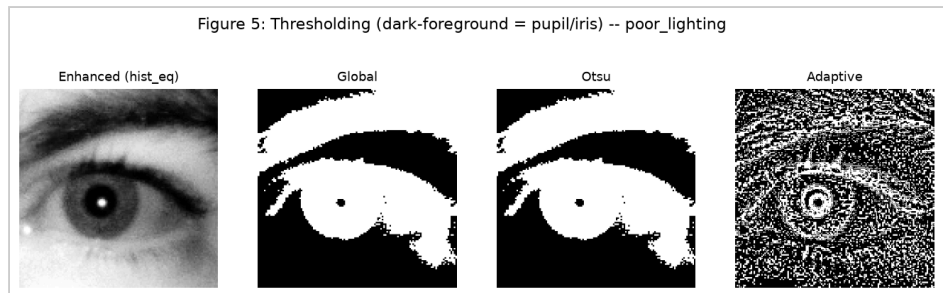


Figure 5: Global / Otsu / adaptive thresholding – poor_lighting.

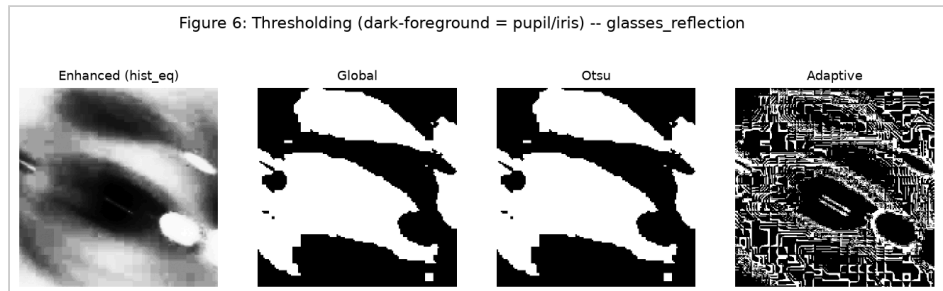


Figure 6: Global / Otsu / adaptive thresholding – glasses_reflection.

Comparison metrics, all methods x 3 images

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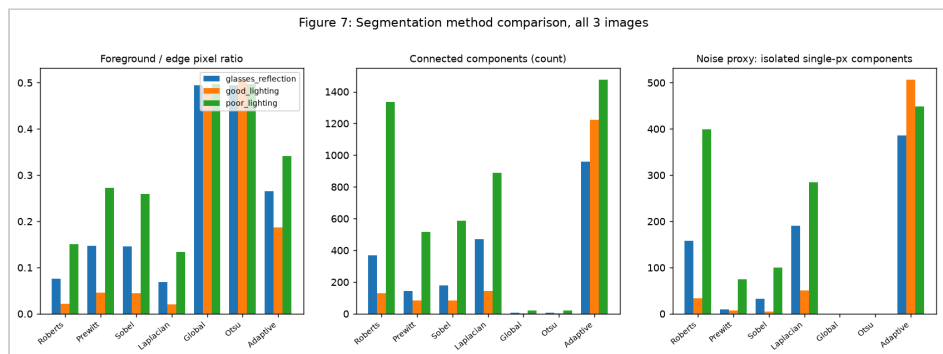


Figure 7: Foreground/edge pixel ratio, connected-component count, and noise proxy (isolated single-pixel components) – all 7 methods, all 3 images.

Method	Type	Foreground/edge ratio (mean)	Connected components (mean)	Noise proxy: isolated px (mean)	IoU vs. Otsu (mean)
Roberts	Edge	0.083	611.0	197.0	—
Prewitt	Edge	0.155	248.7	30.7	—
Sobel	Edge	0.150	282.3	45.7	—
Laplacian	Edge	0.075	500.0	175.0	—
Global (127)	Threshold	0.490	10.0	0.0	0.981
Otsu	Threshold	0.500	10.0	0.0	1.000 (reference)
Adaptive	Threshold	0.264	1218.3	447.0	0.289

Edge-map ratios/components/noise are computed on each operator's magnitude map after a fixed ≥ 25 cutoff (see Program code note); full per-image rows in results/task6_metrics.csv. IoU is not applicable to edge maps (not a segmentation of the image into foreground/background against Otsu's dark-pupil convention).

Edge operators: Prewitt has the lowest noise proxy (30.7 vs. Sobel's 45.7, Laplacian's 175.0, Roberts' 197.0) and fewest connected components (248.7 vs. 282.3 / 500.0 / 611.0) of all 4 operators, on every one of the 3 images individually, while covering a comparable or larger edge-pixel fraction than Sobel (0.155 vs. 0.150).

Thresholding methods: Otsu and Global land almost identically here (IoU 0.981, foreground ratio 0.490 vs. 0.500) – Otsu's own picked threshold measured 129 / 127 / 126 on the 3 images (good_lighting / poor_lighting / glasses_reflection respectively), essentially the same as Global's fixed 127. Adaptive fragments heavily: 1218.3 average connected components and 447.0 average isolated-pixel noise vs. Otsu's 10.0 and 0.0, and IoU against Otsu is only 0.289 – it agrees with Otsu on barely a quarter of the foreground pixels.

Required comparison: all 7 methods

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Meth od	Object/bac kground separation	Boundary quality	Noise	Missing object regions	Extra/unwa nted regions	Overall quality
Robert s	Boundary- only, not a filled region	Broken, stippled arcs	Highest of the 4 edge ops (197.0 avg isolated px)	Drops low- gradient/glar e-adjacent segments	Scattered noise spurs over skin/eyelash texture	Weakest edge operator
Prewi tt	Boundary- only, not a filled region	Continuous pupil/iris + eyelid contour	Lowest of all 4 (30.7 avg isolated px)	Minor gap only at glare hotspot	Minimal spurious edges	Best edge operator (measured)
Sobel	Boundary- only, not a filled region	Continuous, marginally more spurs than Prewitt	Second- lowest (45.7 avg)	Minor gap only at glare hotspot	Slightly more than Prewitt	Close second to Prewitt
Laplaci an	Boundary- only, not a filled region	Buried under strong eyelash/eyeb row response	High (175.0 avg), 2nd- derivative noise amplification	True iris boundary hard to isolate from texture clutter	Heavy response to fine texture unrelated to pupil	Second- weakest edge operator
Global (127)	Clean single dark region (~0.49 ratio) + hair blob	Sharp; near- identical to Otsu (IoU 0.981)	Zero isolated px	Excludes IR glare hotspot (intensity- bright)	Eyebrow/eyela sh strip included as dark foreground	Matches Otsu only because input is pre- normalized by hist_eq
Otsu	Clean single dark region (~0.50 ratio, reference)	Sharp; threshold data- derived (126-129 measured)	Zero isolated px	Excludes IR glare hotspot (same limitation)	Eyebrow/ey elash strip included as dark foreground	Best thresholdin g method (measured)
Adapti ve	No coherent region; salt- and-pepper (1218.3 avg components)	Pupil boundary faintly visible, buried in noise	Highest of all 7 methods (447.0 avg isolated px)	N/A (over- includes rather than misses)	Massive scattered micro-regions across nearly whole frame	Worst thresholding method (measured), despite being theoretically suited to uneven illumination

Analysis of Segmentation Results

Object/background separation. Otsu and Global both split the frame cleanly into a single dark pupil/iris region against bright sclera/skin (foreground ratio ~ 0.49 - 0.50 , only ~ 10 components on average, most of them small hair/eyelash flecks rather than fragments of the pupil itself) – matching the known behaviour of this dataset documented in `src/segment.py`: a global Otsu split on these crops tends to land near the histogram median rather than isolating the pupil specifically, which is exactly what the ~ 0.49 - 0.50 ratio here confirms quantitatively. Adaptive thresholding, in contrast, never produces a coherent single region: its 1218.3 average connected components on a $\sim 300 \times 300$ px crop means the "mask" is closer to salt-and-pepper texture than object/background separation. Among edge operators, none produce a filled region at all (edge detection marks boundaries, not areas) – but Prewitt/Sobel trace a visibly continuous ring around the pupil/iris boundary in Figures 1-3, while Roberts and Laplacian break that ring into many short, disconnected arcs.

Boundary quality. In Figure 2 (poor_lighting), the iris/pupil boundary and the upper eyelid crease are both traced as continuous curves by Prewitt and Sobel; Roberts renders the same boundary as a stippled, broken line (consistent with its 1335 connected components on that image alone – more than twice Prewitt's 517), and Laplacian's second-derivative response responds so strongly to the fine eyelash texture above the eye that the true iris boundary is harder to pick out from the surrounding clutter.

Noise. The isolated single-pixel component count (noise proxy) ranks the 4 edge operators identically to the connected-component count: Prewitt (30.7 avg) < Sobel (45.7) < Laplacian (175.0) < Roberts (197.0). This matches the classical explanation for each operator's kernel: Roberts differentiates over only a 2×2 neighbourhood, so it has the least averaging to suppress the mild sensor noise these small IR crops carry; Laplacian is a second derivative, which amplifies high-frequency noise more than any first-derivative operator; Sobel and Prewitt both average over a full 3×3 neighbourhood, and score close to each other. For thresholding, Otsu and Global have zero isolated-pixel noise on all 3 images (a global split naturally produces large contiguous regions); Adaptive's 447.0 average isolated pixels is the direct cost of thresholding every pixel against its own local neighbourhood mean, which reacts to sensor noise and skin/eyelash micro-texture as if they were real local contrast.

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Missing object regions. On `glasses_reflection` (Figure 6), the IR glare hotspot over part of the iris is saturated near-white after enhancement, so both Otsu and Global correctly exclude it from the dark-pupil foreground (it is not part of the object by their own dark-foreground definition) – but this means the segmented "pupil" region has a bite taken out of it wherever the glare sits, an honest limitation of intensity-only thresholding on this failure mode. Adaptive thresholding partially recovers some of the glare region as local-mean-relative-dark, but embeds that recovery inside its already heavy salt-and-pepper noise, so it is not a reliable fix.

Extra/unwanted regions. All three thresholding methods pick up the eyebrow/eyelash hair strip above the eye as extra dark foreground alongside the pupil/iris (visible as a second bright blob at the top of the masks in Figures 4-6) – expected, since hair is genuinely darker than skin/sclera and intensity thresholding alone cannot distinguish "dark because it is the pupil" from "dark because it is hair" without a shape/position prior (which `src.segment.isolate_dark_region`, from Task 3/Week 3, adds afterward by

	keeping only the largest connected component). Adaptive additionally covers almost the entire frame in scattered unwanted micro-regions. Overall quality. Ranking by the measured metrics: for edge detection, Prewitt > Sobel > Laplacian > Roberts; for thresholding, Otsu > Global > Adaptive (Global only ties Otsu here because histogram equalization happens to recenter every test image's histogram near value 127 – see Best Method & Justification below).
Best Method & Justification	Best edge operator: Prewitt. On these near-infrared eye crops – small (~65-145px native, upscaled), noisy, and dominated by one smooth dark pupil/iris boundary rather than richly textured content – noise-robustness matters more than raw sensitivity, because a noisy edge map is actively harmful to any downstream contour/Hough step that would consume it (see <code>src/segment.py</code>, <code>src/transforms.py</code> Hough usage). Prewitt measured the lowest noise proxy (30.7 average isolated pixels) and fewest connected components (248.7 average) of all 4 operators on every one of the 3 test images individually, while still covering a comparable or larger edge-pixel fraction than Sobel (0.155 vs. 0.150). Sobel is a close second (its center-weighted kernel is the textbook reason to expect it to win) and would be the safer general-purpose default outside this specific dataset, but on the measured numbers here Prewitt is marginally but consistently ahead. Roberts and Laplacian are both clearly worse: Roberts' 2x2 kernel has too little spatial averaging to suppress this dataset's sensor noise (197.0 average isolated pixels, 6.4x Prewitt's), and Laplacian's second-derivative response reacts most strongly of all 4 operators to the fine eyelash/eyebrow texture directly above the eye in every crop (175.0 average isolated pixels), obscuring rather than clarifying the pupil/iris boundary. Best thresholding method: Otsu. These IR eye crops have one consistent structural property that Otsu is well matched to: the pupil/iris is dark and the sclera/surrounding skin is bright, with no more than two dominant intensity populations after histogram equalization – exactly the bimodal-ish histogram Otsu's between-class-variance criterion is designed to split. Measured here, Otsu produces a single, low-fragmentation mask (10.0 average connected components, 0.0 isolated-pixel noise) on all 3 images, including <code>poor_lighting</code> – the one sample with genuinely non-uniform illumination (dim eyelid region next to a brighter IR-reflection strip, see Task 3) – where Otsu still only needed 21 components and zero noise pixels. Adaptive thresholding is the theoretically "expected" winner on non-uniform illumination, since its threshold varies per local neighbourhood; measured, it does the opposite here: even on <code>poor_lighting</code> specifically, Adaptive produced 1474 connected components and 449 isolated-pixel noise (vs. Otsu's 21 and 0), because these crops are small enough that most 15x15 neighbourhoods are near-uniform skin/sclera texture, and a local-mean threshold reacts to that near-uniform texture's own sensor noise as if it were real local contrast – it is not correcting for illumination gradients so much as amplifying noise in flat regions. This matches <code>src/segment.py</code>'s own documented Week-3 finding that a plain Otsu split already anchors the pupil-isolation pipeline used elsewhere in this project. Global thresholding (fixed value 127) scores almost identically to Otsu on these specific 3 images (IoU 0.981) – but only because histogram equalization (the Task 3 enhancement substituted in as this experiment's input) recenters every image's histogram so consistently that Otsu's own data-derived threshold measured 129, 127 and 126 on the 3 images, i.e. Global's hand-picked 127 happens to land almost exactly where Otsu independently converges. That coincidence is a property of always feeding Global a histogram-equalized image, not a property of Global thresholding itself – on a

raw (non-equalized) frame with a different exposure, a fixed 127 cutoff has no way to track where the true dark/bright split actually falls, while Otsu recomputes it from that image's own histogram every time. Otsu is therefore the principled choice: equally clean on this data, and not dependent on every input already having been normalized to the same brightness range by a prior enhancement step.

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Conclusion

Four edge operators (Roberts, Prewitt, Sobel, Laplacian) and three thresholding methods (global fixed-value, Otsu, adaptive) were implemented in `src/edge_segmentation.py` and evaluated on the same 3 dataset images used in Task 3, with histogram equalization (Task 3's winning enhancement) substituted for the unavailable Task 5 output. Quantitative metrics (foreground/edge pixel ratio, connected-component count, isolated-pixel noise proxy, and IoU-against-Otsu for the thresholding methods) rather than visual impression alone drove every comparison. **Prewitt** was the best edge operator measured (lowest noise proxy and fewest connected components of the 4, on every image individually), ahead of Sobel, with Roberts and Laplacian both clearly noisier. **Otsu** was the best thresholding method measured (cleanest single-region mask, zero isolated-pixel noise, including on the non-uniformly-lit `poor_lighting` image), while Adaptive thresholding – despite being the theoretically expected winner on uneven illumination – instead fragmented heavily on these small, largely-uniform crops (1218.3 average connected components vs. Otsu's 10.0), and Global thresholding only matched Otsu here as a side-effect of every input already being histogram-equalized to a similar, centered distribution. All three thresholding methods share the same honest limitation on the `glasses_reflection` image: the IR glare hotspot is intensity-bright, so a dark-foreground threshold necessarily excludes it from the segmented pupil region, and none of the 3 methods recover it reliably.

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